# Silent Voice Bridge: The Edge AI ASL Interpreter

# Report 3: Testing and Evaluation Plan

**CMPE 195A - Senior Design Project I Spring 2026**

**Team Members:**

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**Page Limit:** Sections 1–3 below must fit within **1 page**. This cover and the References section do not count.

**Feel free to remove all the red text it is just there for instructions.**

# 1. Testing Strategy

Describe the testing approaches your team will use and the tools that support them. Focus on the methods most relevant to your project type.

*[2–4 sentences or bullet points. Example: "We will use unit testing with Jest to validate individual components, and conduct user testing with 5–10 participants to gather usability feedback. Load testing will be performed using Locust to verify the system handles 100 concurrent users."]*

- **Unit testing** with pytest to make sure individual components work (MediaPipe hand landmark extraction, LSTM inference wrapper, frame preprocessing pipeline).
- **Model evaluation** by testing the trained model on ASL videos that it has not seen before, to measure how often it correctly recognizes the sign.
- **Latency/performance testing** on the Jetson hardware to make sure the system is fast enough for real-time translation (target: <300ms per sign for usable real-time translation).
- **User testing** with 5–10 participants signing a set of ASL words to see how well the device works with different people, lighting, and camera angles.

# 2. Evaluation Criteria

List the specific, measurable metrics that will determine whether your project is successful. Each criterion should include a target value and how it will be measured.

- **[Metric 1]:** Target — [value]. Measured by — [method].
- **[Metric 2]:** Target — [value]. Measured by — [method].
- **[Metric 3]:** Target — [value]. Measured by — [method].

- [Real-time Latency]: Target - Under 300ms. Measured by - Timing the communication pipeline between the Python backend and the React frontend.
- [Processing Speed]: Target - Minimum of 30 FPS. Measure by - Deploying the MediaPipe hand tracking pipeline on the NVIDIA Jetson and monitoring frame processing rates.
- [Static Recognition Accuracy]: Target - High accuracy for the static ASL alphabet (A-Z) and 5-10 static words. Measured by - Testing the baseline Random Forest Classifier against the Kaggle ASL Alphabet dataset.
- [Full Vocabulary Accuracy]: Target - 85% or higher. Measured by - Evaluating the

model's performance on a curated dictionary of 50+ common conversational ASL phrases in varied lighting conditions.

- [Linguistic Scope]: Target - Recognition of dynamic, word-level signs. Measured by - Integrating LSTM-based/ A.I. Vision model and testing with the WLASL dataset.

## 3. Testing Timeline (CMPE 195B)

Briefly outline when each major testing phase will take place during 195B.

- **Early (Weeks 1–4):** [Testing activities]
- **Mid (Weeks 5–10):** [Testing activities]
- **Late (Weeks 11–14):** [Testing activities]
- **Final (Weeks 15–16):** [Final validation before Expo]

Weeks 1-4:

- Very camera integration on Jetson
- Test dataset and processing pipeline
- Train an AI model (baseline ASL)
- Initial inference testing on Jetson (without optimization)
- Test accuracy, inference time per frame

Weeks 5-10:

- Deploy model using ONNX with optimization with TensorRT on Jetson
- Benchmark performance improvements
- Check accuracy under different conditions

Weeks 11-14:

- Integrate all subsystems (camera + jetson + output interface)
- Stress testing
- End-to-end testing
- Test edge cases
- Refine the model based on test results

Weeks 15-16:

- Full system validation for demo
- More testing for consistent real-time performance
- Final testing on accuracy, latency, and usability
- Prepare a demo for the presentation